

Medical Q&A Chatbot Using NLP

Abstract

A Medical Q&A Chatbot is an AI-based system that provides answers to medical questions using Natural Language Processing. The system uses the MedQuAD dataset and machine learning techniques to find the most relevant answers.

Objectives

- Develop an AI healthcare assistant
- Process medical questions using NLP
- Retrieve accurate answers from dataset
- Provide user-friendly web interface

Methodology

1. Collect medical question-answer dataset
2. Clean and preprocess text
3. Convert questions into TF-IDF vectors
4. Calculate similarity using cosine similarity
5. Display the best matching answer

System Architecture

User → Streamlit → Chatbot → NLP Model → Dataset → Answer

Tools Used

Python

Streamlit

Scikit-learn

Pandas

NLP

Results

The chatbot successfully retrieves relevant medical answers based on user queries.

(Add your screenshots here)

Future Scope

- Voice assistant
- Deep learning models
- Real-time healthcare integration

Conclusion

The project shows the application of AI and NLP in healthcare support systems.